

Airflow - Two operators

The DockerOperator:

Grant read/write on the Docker socket

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ sudo chmod a+rw /var/run/docker.sock
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$
```

Edit docker-compose.yaml file to include

- environment: _PIP_ADDITIONAL_REQUIREMENTS:
\${_PIP_ADDITIONAL_REQUIREMENTS:- apache-airflow-providers-docker}
- volume: /var/run/docker.sock:/var/run/docker.sock

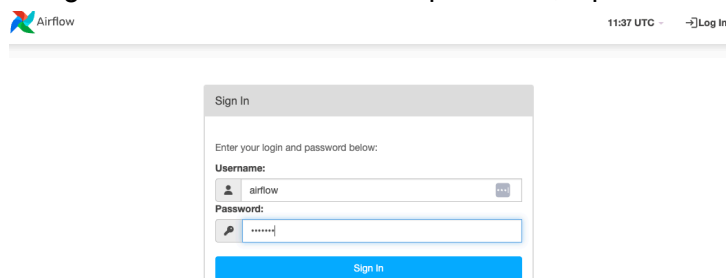
Recreate containers

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker-compose up --force-recreate -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Recreating airflow_redis_1 ... done
Recreating airflow_postgres_1 ... done
Recreating airflow_airflow-scheduler_1 ... done
Recreating airflow_airflow-init_1 ... done
Recreating airflow_flower_1 ... done
Recreating airflow_airflow-webserver_1 ... done
Recreating airflow_airflow-worker_1 ... done
```

Test DockerOperator with “hello-world”, create dags/hello_world.py file

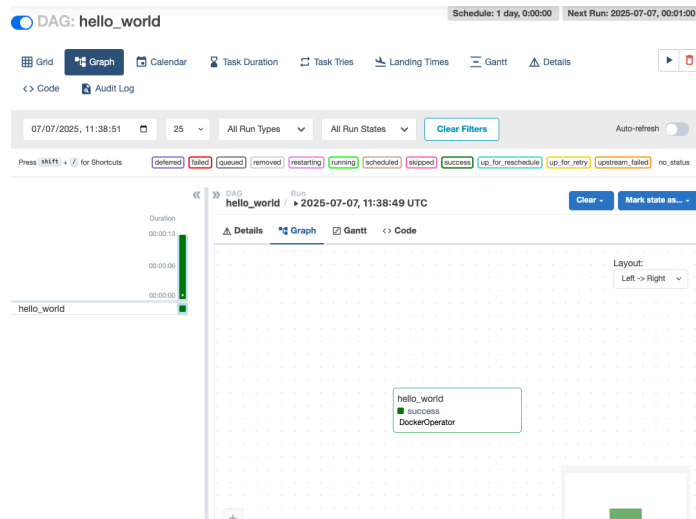
```
hello_world.py U X
datascientest-data-engineering > airflow > dags > 📄 hello_world.py
1 from airflow import DAG
2 from airflow.utils.dates import days_ago
3 from airflow.operators.docker_operator import DockerOperator
4
5
6 with DAG(
7     dag_id = "hello_world",
8     tags = ["docker", "datascientest"],
9     default_args = {
10         "owner": "airflow",
11         "start_date": days_ago(0, minute=1),
12     },
13     catchup = False
14 ) as dag:
15     hello_world = DockerOperator(
16         task_id = "hello_world",
17         image = "hello-world"
18     )
```

Using VM IP 34.252.69.172 and port 8080, Open Airflow UI <http://34.252.69.172:8080/>



The screenshot shows the Airflow web interface. At the top, there is a header with the Airflow logo, the time "11:37 UTC", and a "Log In" link. Below the header is a "Sign In" form. The form has a title "Sign In" and a subtitle "Enter your login and password below:". It contains two input fields: "Username:" with the value "airflow" and "Password:" with masked characters. A blue "Sign In" button is at the bottom of the form.

Trigger hello_world DAG



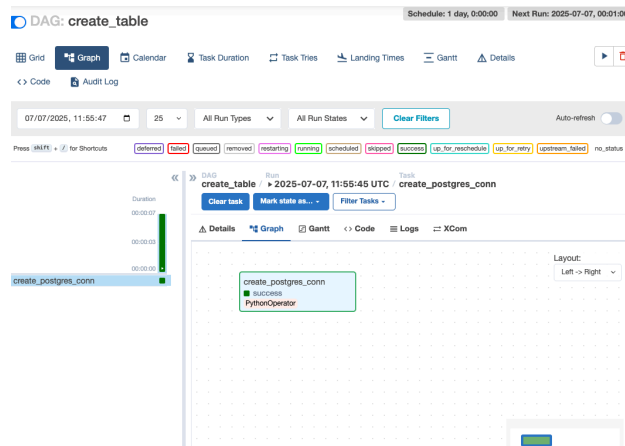
Open terminal in VM to verify

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/dags$ docker ps -a | grep hello-world
a98c7714f32b    hello-world    "/hello"          5 minutes ago    Exited (0) 5 minutes ago
intelligent_goldstine
4e1372735475    hello-world    "/hello"          4 years ago      Exited (0) 4 years ago
festive_elgamal
```

The PostgresOperator: Add PostgresOperator and PythonOperator for create_table. Create dags/create_table.py file

```
create_table.py U X
datascientest-data-engineering > airflow > dags > create_table.py
1 from airflow import DAG, settings
2 from airflow.utils.dates import days_ago
3 from airflow.operators.python_operator import PythonOperator
4 from airflow.models.connection import Connection
5
6
7 def create_postgres_conn(**kwargs):
8     session = settings.Session()
9     print("Session created")
10    connections = session.query(Connection)
11    print("Connections listed")
12    if not kwargs["conn_id"] in [connection.conn_id for connection in connections]:
13        conn = Connection(
14            conn_id = kwargs["conn_id"],
15            conn_type = "postgres",
16            host = "postgres",
17            login = "airflow",
18            password = "airflow",
19            schema = "airflow"
20        )
21        session.add(conn)
22        session.commit()
23        print("Connection Created")
24    else:
25        print("Connection already exists")
26    session.close()
27
28 with DAG(
29     dag_id = "create_table",
30     tags = ["postgres", "datascientest"],
31     default_args = {
32         "owner": "airflow",
33         "start_date": days_ago(0, minute=1),
34     },
35     catchup = False
36 ) as dag:
37     create_conn = PythonOperator(
38         task_id = "create_postgres_conn",
39         python_callable = create_postgres_conn,
40         op_kwargs = {"conn_id": "postgres"}
41     )
```

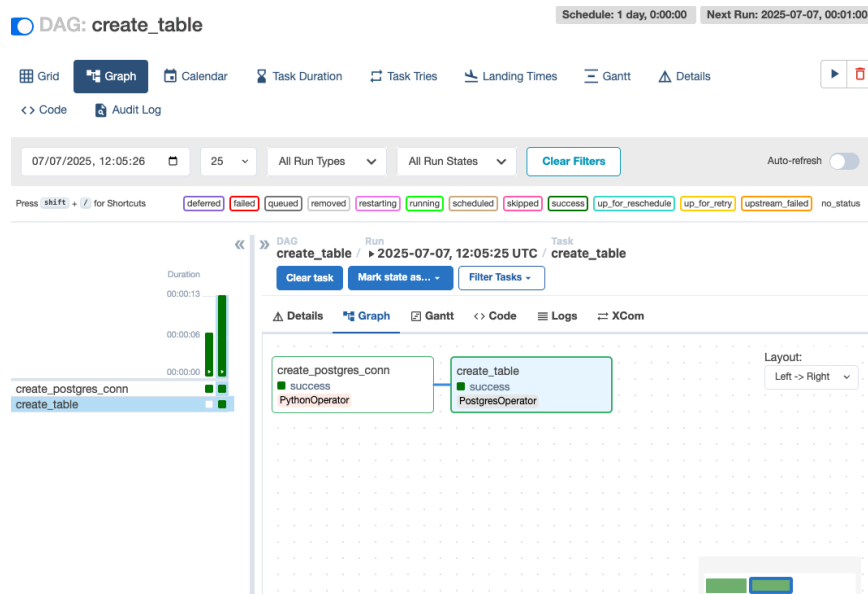
Open Airflow UI, trigger create_postgres_conn to create the connection



Update dags/create_table.py file

```
create_table.py U X
tascentest-data-engineering > airflow > dags > create_table.py
1 from airflow.operators.postgres_operator import PostgresOperator
2 from airflow import DAG, settings
3 from airflow.utils.dates import days_ago
4 from airflow.operators.python_operator import PythonOperator
5 from airflow.models.connection import Connection
6
7
8 def create_postgres_conn(**kwargs):
9     session = settings.Session()
10    print("Session created")
11    connections = session.query(Connection)
12    print("Connections listed")
13    if not kwargs["conn_id"] in [connection.conn_id for connection in connections]:
14        conn = Connection(
15            conn_id = kwargs["conn_id"],
16            conn_type = "postgres",
17            host = "postgres",
18            login = "airflow",
19            password = "airflow",
20            schema = "airflow"
21        )
22        session.add(conn)
23        session.commit()
24        print("Connection Created")
25    else:
26        print("Connection already exists")
27    session.close()
28
29 with DAG(
30     dag_id = "create_table",
31     tags = ["postgres", "datascientest"],
32     default_args = {
33         "owner": "airflow",
34         "start_date": days_ago(0, minute=1),
35     },
36     catchup = False
37 ) as dag:
38     create_conn = PythonOperator(
39         task_id = "create_postgres_conn",
40         python_callable = create_postgres_conn,
41         op_kwargs = {"conn_id": "postgres"}
42     )
43
44     create_table = PostgresOperator(
45         task_id = "create_table",
46         postgres_conn_id = "postgres",
47         sql = """
48         CREATE TABLE IF NOT EXISTS customer (
49             id UUID PRIMARY KEY,
50             lastname VARCHAR NOT NULL,
51             firstname VARCHAR NOT NULL,
52             sex VARCHAR NOT NULL,
53             street_number SMALLINT NOT NULL,
54             street_name VARCHAR NOT NULL,
55             city VARCHAR NOT NULL,
56             postcode VARCHAR NOT NULL,
57             region VARCHAR NOT NULL,
58             modification_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
59         """
60     )
61     create_conn >> create_table
```

Open Airflow UI, trigger create_table to create the customer table



Exercise:

Build own two-DAG exercise

Create a sql subfolder under airflow/dags/ and add three new files:

- create_table_customer.sql
- create_table_product.sql
- create_table_order.sql

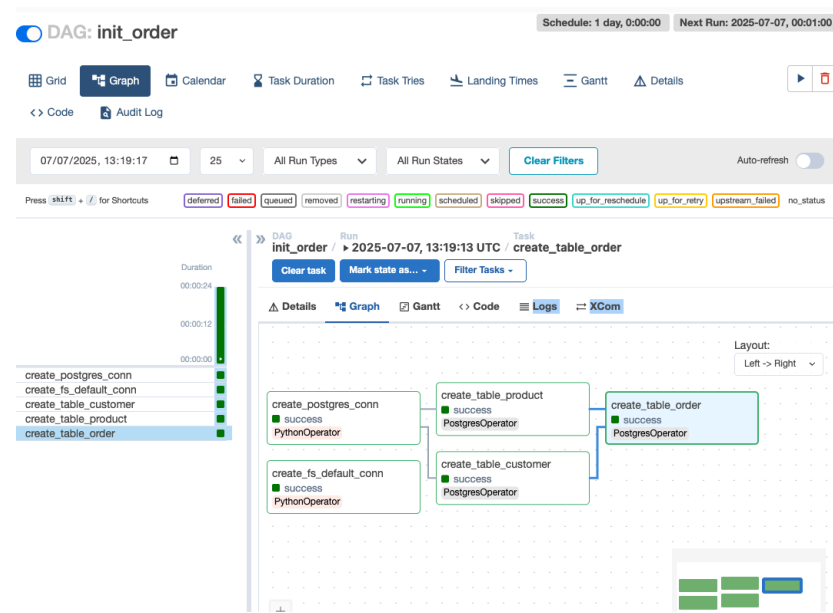
```
create_table_customer.sql U X
atascientest-data-engineering > airflow > dags > sql > create_table_
1 CREATE TABLE IF NOT EXISTS customer (
2   id UUID PRIMARY KEY,
3   lastname VARCHAR,
4   firstname VARCHAR,
5   sex VARCHAR,
6   street_number SMALLINT,
7   street_name VARCHAR,
8   city VARCHAR,
9   postcode VARCHAR,
10  region VARCHAR,
11  modification_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
12 );
```

```
create_table_product.sql U X
atascientest-data-engineering > airflow > dags > sql > create_table_p
1 CREATE TABLE IF NOT EXISTS product (
2   id UUID PRIMARY KEY,
3   name VARCHAR,
4   categories VARCHAR,
5   price REAL,
6   weight REAL,
7   modification_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
8 );
```

```
create_table_order.sql U X
atascientest-data-engineering > airflow > dags > sql >
1 CREATE TABLE IF NOT EXISTS "order" (
2   id UUID,
3   date_order TIMESTAMP,
4   date_shipping TIMESTAMP,
5   quantity SMALLINT,
6   price REAL,
7   customer_id UUID REFERENCES customer(id),
8   product_id UUID REFERENCES product(id)
9 );
```

For first DAG init create `airflow/dags/init_order.py`

In Airflow UI trigger init_order DAG



Open terminal in VM to verify

```
warning-172-31-46-112: /etc/cni/netconf/airflow/docker-compose up airflow-init
WARNING: The ALERT_MAIL variable is not set. Defaulting to a blank string.
Building with native build. Learn about native build in Compose here: https://docs.docker.com/compose-native-build/
airflow_postgres_1 is up-to-date
airflow_postgres_1 is up-to-date
Creating airflow_airflow-init_1 ... done
Attaching to airflow_airflow-init_1
airflow-init-1
airflow-init-1 WARNING: You should run the image with GID (Group ID) set to 0
airflow-init-1 even if you use "airflow" user (UID=5000)
airflow-init-1
airflow-init-1 You started the image with UID=50000 and GID=50000
airflow-init-1
airflow-init-1 This is to make sure you can run the image with an arbitrary UID in the future.
airflow-init-1
airflow-init-1 See more about it in the Airflow's docker image documentation
airflow-init-1 http://airflow.apache.org/docs/docker-stack/entrypoint
airflow-init-1
airflow-init-1
/home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:793 FutureWarning: The auth_backends setting in [api] has had airflow_api_auth_backend.session added in the running config, which is needed by the UI. Please update te your config before Apache Airflow 3.0.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please u
update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please
update your config.
airflow-init-1 DB: postgresql+psycopg2://airflow@postgres/airflow
airflow-init-1 Performing upgrade to the metadata database postgresql+psycopg2://airflow@postgres/airflow
[2025-07-07T13:24:21.156+0000] [migration.py:216] INFO - Context impl PostgresqlImpl.
[2025-07-07T13:24:21.156+0000] [migration.py:219] INFO - Will assume transactional DDL.
[2025-07-07T13:24:21.295+0000] [db.py:1616] INFO - Creating tables
airflow-init-1 INFO - Scheduling runtime migration will assume transactional DDL.
airflow-init-1 INFO - Scheduling runtime migration will assume transactional DDL.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/models/dagbag.py:344 RemovedInAirflowWarning: Param 'schedule_interval' is deprecated and will be removed in a future release. Please use 'schedule' instead.
airflow-init-1 Database migrating done!
airflow-init-1 WARNING: Environment variable `AIRFLOW_DB_UPGRADE` is deprecated please use `AIRFLOW_DB_MIGRATE` instead
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please
update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please update your config.
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airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:761 FutureWarning: The auth_backends setting in [api] has had airflow_api_auth_backend.session added in the running config, which is needed by the UI. Please upd
te your config before Apache Airflow 3.0.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please
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update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please
update your config.
airflow-init-1 /home/airflow/local/lib/python3.8/site-packages/flask_limiter/extensions.py:338 UserWarning: Using the in-memory storage for tracking rate limits as no storage was explicitly specified. This is not recommended for production use. See h
ttps://flask-limiter.readthedocs.io/en/configuring-storage-backend-for-documentation about configuring the storage backend.
airflow-init-1 airflow already exist in the db
airflow-init-1
airflow-init-1 !!!!! Installing additional requirements: 'python-pgcrypto-airflow-providers-docker' !!!!!!!!!!!!!!!
airflow-init-1
airflow-init-1 WARNING: This is a development/test feature only. NEVER use it in production!
airflow-init-1 Instead, build a custom image as described in
airflow-init-1 https://airflow.apache.org/docs/docker-stack/build.html
airflow-init-1
airflow-init-1 Adding requirements at container startup is fragile and is done every time
airflow-init-1 the container starts, so it is only useful for testing and trying out
airflow-init-1 of various dependencies.
```

```

airflow-init_1 | Requirement already satisfied: opentelemetry-sdk==1.22.0 in /home/airflow/.local/lib/python3.8/site-packages (from opentelemetry-exporter-otlp-proto-grpc==1.22.0->opentelemetry-e
xporter-otlp->apache-airflow==2.6.0->apache-airflow-providers-docker) (1.22.0)
airflow-init_1 | Requirement already satisfied: protobuf<5.0,>=3.19 in /home/airflow/.local/lib/python3.8/site-packages (from opentelemetry-proto==1.22.0->opentelemetry-exporter-otlp-proto-grpc=
1.22.0->opentelemetry-exporter-otlp->apache-airflow==2.6.0->apache-airflow-providers-docker) (4.25.2)
airflow-init_1 | Requirement already satisfied: pycparser in /home/airflow/.local/lib/python3.8/site-packages (from cffi==1.12->cryptography==0.9.3->apache-airflow==2.6.0->apache-airflow-provider
s-docker) (2.21)
airflow-init_1 | Requirement already satisfied: Babel==2.3 in /home/airflow/.local/lib/python3.8/site-packages (from Flask-Babel<3,>=1->flask-appbuilder==4.3.10->apache-airflow==2.6.0->apache-air
flow-providers-docker) (2.14.0)
airflow-init_1 | Requirement already satisfied: limits>=2.8 in /home/airflow/.local/lib/python3.8/site-packages (from Flask-Limiter<4,>3->flask-appbuilder==4.3.10->apache-airflow==2.6.0->apache-a
irflow-providers-docker) (3.7.0)
airflow-init_1 | Requirement already satisfied: ordered-set<5,>4 in /home/airflow/.local/lib/python3.8/site-packages (from Flask-Limiter<4,>3->flask-appbuilder==4.3.10->apache-airflow==2.6.0->apa
che-airflow-providers-docker) (4.1.0)
airflow-init_1 | Requirement already satisfied: h11<0.15,>=0.13 in /home/airflow/.local/lib/python3.8/site-packages (from httpcore<0.17.0,>=0.15.0->httpx->apache-airflow==2.6.0->apache-airflow-pr
oviders-docker) (0.14.0)
airflow-init_1 | Requirement already satisfied: anyio<5.0,>=3.0 in /home/airflow/.local/lib/python3.8/site-packages (from httpcore<0.17.0,>=0.15.0->httpx->apache-airflow==2.6.0->apache-airflow-pr
oviders-docker) (4.2.0)
airflow-init_1 | Requirement already satisfied: multidict<7.0,>=4.5 in /home/airflow/.local/lib/python3.8/site-packages (from aiohttp->apache-airflow-providers-http->apache-airflow==2.6.0->apache-a
irflow-providers-docker) (6.0.4)
airflow-init_1 | Requirement already satisfied: yarl<2.0,>=1.0 in /home/airflow/.local/lib/python3.8/site-packages (from aiohttp->apache-airflow-providers-http->apache-airflow==2.6.0->apache-airf
low-providers-docker) (1.9.4)
airflow-init_1 | Requirement already satisfied: frozenlist==1.1.1 in /home/airflow/.local/lib/python3.8/site-packages (from aiohttp->apache-airflow-providers-http->apache-airflow==2.6.0->apache-a
irflow-providers-docker) (1.4.1)
airflow-init_1 | Requirement already satisfied:aiosignal>=1.1.2 in /home/airflow/.local/lib/python3.8/site-packages (from aiohttp->apache-airflow-providers-http->apache-airflow==2.6.0->apache-a
irflow-providers-docker) (1.3.1)
airflow-init_1 | Requirement already satisfied: async-timeout<5.0,>=4.0 in /home/airflow/.local/lib/python3.8/site-packages (from aiohttp->apache-airflow-providers-http->apache-airflow==2.6.0->ap
ache-airflow-providers-docker) (4.0.3)
airflow-init_1 | Requirement already satisfied: exceptiongroup>=1.0.2 in /home/airflow/.local/lib/python3.8/site-packages (from anyio<5.0,>=3.0->httpcore<0.17.0,>=0.15.0->httpx->apache-airflow==2
.6.0->apache-airflow-providers-docker) (1.2.0)
airflow-init_1 | Requirement already satisfied: opentelemetry-semantic-conventions==0.43b0 in /home/airflow/.local/lib/python3.8/site-packages (from opentelemetry-sdk==1.22.0->opentelemetry-expor
ter-otlp-proto-grpc==1.22.0->opentelemetry-exporter-otlp->apache-airflow==2.6.0->apache-airflow-providers-docker) (0.43b0)
airflow-init_1 | Downloading pymongo-4.10.1-cp38-cp38-manylinux_2_17_x86_64_manylinux2014_x86_64.whl (530 kB)
airflow-init_1 | ----- 930.2/930.2 kB 8.1 MB/s eta 0:00:00
airflow-init_1 | Installing collected packages: pymongo
airflow-init_1 | Successfully installed pymongo-4.10.1
airflow-init_1 | [notice] A new release of pip is available: 23.3.2 -> 25.0.1
airflow-init_1 | [notice] To update, run: python -m pip install --upgrade pip
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn opt
ion in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old set
ting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old set
ting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:761 FutureWarning: The auth_backends setting in [api] has had airflow.api.auth.backend.session added in
the running config, which is needed by the UI. Please update your config before Apache Airflow 3.0.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn opt
ion in [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option i
n [database] - the old setting has been used, but please update your config.
airflow-init_1 | /home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sqlalchemy_conn option in [core] has been moved to the sqlalchemy_conn option
in [database] - the old setting has been used, but please update your config.
airflow-init_1 | 2.8.1
airflow-airflow-init_1 exited with code 0

```

Log into Postgres in order to ensure the tables exist

```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/dags$ docker exec -it airflow_postgres_1 \
> psql -U airflow -d airflow \
> -c "\dt"

```

Schema	Name	Type	Owner
public	ab_permission	table	airflow
public	ab_permission_view	table	airflow
public	ab_permission_view_role	table	airflow
public	ab_register_user	table	airflow
public	ab_role	table	airflow
public	ab_user	table	airflow
public	ab_user_role	table	airflow
public	ab_view_menu	table	airflow
public	alembic_version	table	airflow
public	callback_request	table	airflow
public	celery_taskmeta	table	airflow
public	celery_tasksetmeta	table	airflow
public	connection	table	airflow
public	customer	table	airflow
public	dag	table	airflow
public	dag_code	table	airflow
public	dag_owner_attributes	table	airflow
public	dag_pickle	table	airflow
public	dag_run	table	airflow
public	dag_run_note	table	airflow
public	dag_schedule_dataset_reference	table	airflow
public	dag_tag	table	airflow
public	dag_warning	table	airflow
public	dagrun_dataset_event	table	airflow
public	dataset	table	airflow
public	dataset_dag_run_queue	table	airflow
public	dataset_event	table	airflow
public	import_error	table	airflow
public	job	table	airflow
public	log	table	airflow
public	log_template	table	airflow
public	order	table	airflow
public	product	table	airflow
public	rendered_task_instance_fields	table	airflow
public	serialized_dag	table	airflow
public	session	table	airflow
public	sla_miss	table	airflow
public	slot_pool	table	airflow
public	task_fail	table	airflow
public	task_instance	table	airflow
public	task_instance_note	table	airflow
public	task_map	table	airflow
public	task_outlet_dataset_reference	table	airflow
public	task_reschedule	table	airflow
public	trigger	table	airflow
public	variable	table	airflow
public	xcom	table	airflow

(47 rows)

Prepare the raw data

Create a data folder in the airflow folder and upload the order data to the airflow/data folder

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir data
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/orders.tar -P data
--2025-07-07 13:39:56-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/orders.tar
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 52.95.156.94, 3.5.204.163
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|52.95.156.94|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 460800 (450K) [application/x-tar]
Saving to: 'data/orders.tar'

orders.tar                               100%[=====] 450.00K  --.-KB/s   in 0.09s

2025-07-07 13:39:56 (5.00 MB/s) - 'data/orders.tar' saved [460800/460800]

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ tar xvf data/orders.tar -C data
orders/
orders/2024-05-13.json
orders/2024-05-10.json
orders/2024-05-11.json
orders/2024-05-09.json
orders/2024-05-12.json
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ rm data/orders.tar
```

Build & test the python_transform image. Dev folder and Dockerfile

In airflow folder, create the docker/dev/python_transform folders

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir -p docker/dev/python_transform
```

In docker/dev/python_transform folder, create a Dockerfile file and a requirements.txt file

Dockerfile U X	requirements.txt U X
<pre>datascientest-data-engineering > airflow > docker > dev > 1 FROM python:3.8-slim 2 3 WORKDIR /app 4 5 COPY ./requirements.txt ./requirements.txt 6 7 RUN pip install -r requirements.txt</pre>	<pre>datascientest-data-engineering > airflow 1 numpy==1.24.4 2 pandas==1.5.3 3 python-dateutil==2.9.0.post0 4 pytz==2024.1 5 six==1.16.0</pre>

Create the python_transform Docker image

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker build -t python_transform .
[+] Building 50.5s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 159B
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load metadata for docker.io/library/python:3.8-slim
=> [internal] load build context
=> => transferring context: 125B
=> [1/4] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f
=> resolve docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f
=> sha256:b5f62925bd0f63f48cc8acd5e87d0c3a07e2f229cd2fb0a9586e68ed17f45ee3 5.25kB / 5.25kB
=> sha256:302e3ee498053a7b5332ac79e8efebec16e900289fc1ecd1c754ce8fa047fcab 29.13MB / 29.13MB
=> sha256:030d7bdc20a63e3d22192b292d006a69fa3333949f536d62865d1bd0506685cc 3.51MB / 3.51MB
=> sha256:a3f1dfe736c5f959143f23d75ab522a60be2da902efac236f4fb2a153cc14a5d 14.53MB / 14.53MB
=> sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f 10.41kB / 10.41kB
=> sha256:314bc2fb0714b7807bf5699c98f0c73817e579799f2d91567ab7e9510f5601a5 1.75kB / 1.75kB
=> sha256:3971691a363796c39467aae4cdce6ef773273fe6bfc67154d01e1b589befb912 248B / 248B
=> extracting sha256:302e3ee498053a7b5332ac79e8efebec16e900289fc1ecd1c754ce8fa047fcab
=> extracting sha256:030d7bdc20a63e3d22192b292d006a69fa3333949f536d62865d1bd0506685cc
=> extracting sha256:a3f1dfe736c5f959143f23d75ab522a60be2da902efac236f4fb2a153cc14a5d
=> extracting sha256:3971691a363796c39467aae4cdce6ef773273fe6bfc67154d01e1b589befb912
=> [2/4] WORKDIR /app
=> [3/4] COPY ./requirements.txt ./requirements.txt
=> [4/4] RUN pip install -r requirements.txt
=> exporting to image
=> exporting layers
=> writing image sha256:168e197115b849ef331e002bf32f6620e2b1e29ff2436aebf94f50285aeae38
=> naming to docker.io/library/python_transform:latest
```

Prepare directories for bind-mount

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ mkdir -p app/data/to_ingest/bronze
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ mkdir app/data/to_ingest/silver
```

Populate the transform container's Bronze folder and test it

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker run -it \
> -v $PWD/app:/app \
> --name python_transform_dev \
> python_transform bash
root@9185686c992b:/app#
```

Copy one of the raw order files into the container's /app/data/to_ingest/bronze/orders.json (in second terminal)

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cp data/orders/2024-05-09.json docker/dev/python_transform/app/data/to_ingest/bronze/orders.json
```

Download the transform script as main.py file

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cd docker/dev/python_transform/app/
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform/app$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/transform.py -O main.py
--2025-07-07 14:12:40-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/transform.py
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 3.5.205.184, 3.5.206.186
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|3.5.205.184|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2592 (2.5K) [text/x-python]
Saving to: 'main.py'

main.py                               100%[=====>] 2.53K --.-KB/s in 0s

2025-07-07 14:12:40 (57.3 MB/s) - 'main.py' saved [2592/2592]
```

Rebuild the dev image

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker build -t python_transform .
[+] Building 0.9s (9/9) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 159B 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load metadata for docker.io/library/python:3.8-slim 0.7s
=> [1/4] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a61 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 37B 0.0s
=> CACHED [2/4] WORKDIR /app 0.0s
=> CACHED [3/4] COPY ./requirements.txt ./requirements.txt 0.0s
=> CACHED [4/4] RUN pip install -r requirements.txt 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:168e197115b849ef331e002bf32f6620e2b1e29ff2436aebf94fd50285aeae38 0.0s
=> => naming to docker.io/library/python_transform:latest 0.0s
```

Return to first terminal and launch the container mounting app to /app

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker run -it \
> -v $PWD/app:/app \
> --name python_transform_dev \
> python_transform bash
root@6c74e0f18288:/app#
```

Run the transform inside the container

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker run -it \
> -v $PWD/app:/app \
> --name python_transform_dev \
> python_transform bash
root@6c74e0f18288:/app# python3 main.py
orders, customers and products loaded in data/to_ingest/silver directory
root@6c74e0f18288:/app#
```

```
docker /dev /python_transform
├─ app
├─ data/to_ingest
│   └─ bronze
│       ├── {} orders.json
│       └─ silver
│           ├── customers.csv
│           ├── {} orders.json
│           └── products.csv
└─ main.py
```

Clean up after verification

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker run -it \
> -v $PWD/app:/app \
> --name python_transform_dev \
> python_transform bash
root@6c74e0f18288:/app# python3 main.py
orders, customers and products loaded in data/to_ingest/silver directory
root@6c74e0f18288:/app# exit
exit
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/dev/python_transform$ docker rm python_transform_dev
python_transform_dev
```

Production start-up

Make the data folder visible to all Airflow services

Create silver folder so that Airflow worker may view both bronze (raw) and silver (transformed) data

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir -p data/to_ingest/silver
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir -p data/to_ingest/bronze
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cp data/orders/2024-05-09.json data/to_ingest/bronze/orders
.json
```

Update docker-compose.yaml with line - `./data:/opt/airflow/data`

Recreate the Airflow stack in order for new mount to get picked up

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker-compose up --force-recreate -d
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Recreating airflow_postgres_1 ... done
Recreating airflow_redis_1 ... done
Recreating airflow_airflow-init_1 ... done
Recreating airflow_flowable_1 ... done
Recreating airflow_airflow-worker_1 ... done
Recreating airflow_airflow-webserver_1 ... done
Recreating airflow_airflow-scheduler_1 ... done
```

Build the production Docker images

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mkdir -p airflow/docker/prod
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ wget https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/transform_load_prod.tar \
> -P airflow/docker/prod
--2025-07-07 14:59:21-- https://dst-de.s3.eu-west-3.amazonaws.com/airflow/order_example/transform_load_prod.tar
Resolving dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)... 16.12.18.14, 16.12.20.14
Connecting to dst-de.s3.eu-west-3.amazonaws.com (dst-de.s3.eu-west-3.amazonaws.com)|16.12.18.14|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20480 (20K) [application/x-tar]
Saving to: 'airflow/docker/prod/transform_load_prod.tar'

transform_load_prod.tar      100%[=====] 20.00K  --KB/s   in 0.001s

2025-07-07 14:59:21 (28.7 MB/s) - 'airflow/docker/prod/transform_load_prod.tar' saved [20480/20480]

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ tar xvf airflow/docker/prod/transform_load_prod.tar -C airflow/docker/prod
python_load/
python_load/main.py
python_load/requirements.txt
python_load/Dockerfile
python_transform/
python_transform/main.py
python_transform/requirements.txt
python_transform/.env
python_transform/Dockerfile
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ rm airflow/docker/prod/transform_load_prod.tar
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cd airflow/docker/prod/python_transform
```

```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/airflow/docker/prod/python_transform$ docker build -t python_transform .
[+] Building 52.0s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 216B
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load metadata for docker.io/library/python:3.8-slim
=> [1/6] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f
=> [internal] load build context
=> => transferring context: 2.75kB
=> CACHED [2/6] WORKDIR /app
=> [3/6] COPY ./main.py ./main.py
=> [4/6] COPY ./requirements.txt ./requirements.txt
=> [5/6] RUN mkdir -p data/to_ingest
=> [6/6] RUN pip install -r requirements.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:4cf3a198f860a3558bc3818665a0344d4fc50ed1dda428c4f15de20cc4825d63
=> => naming to docker.io/library/python_transform:latest
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/airflow/docker/prod/python_transform$ cd ../python_load
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/airflow/docker/prod/python_load$ docker build -t python_load .
[+] Building 166.5s (12/12) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 282B
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load metadata for docker.io/library/python:3.8-slim
=> CACHED [1/7] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a613f
=> [internal] load build context
=> => transferring context: 4.67kB
=> [2/7] RUN apt-get update -y && apt-get install -y libpq-dev gcc
=> [3/7] WORKDIR /app
=> [4/7] RUN mkdir -p data/to_ingest/silver
=> [5/7] COPY ./main.py ./main.py
=> [6/7] COPY ./requirements.txt ./requirements.txt
=> [7/7] RUN pip install -r requirements.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:ca21d078dc60329f4f09039f6c89dd65536230cc2d422d37dfcadc2633883d33
=> => naming to docker.io/library/python_load:latest

```

Correct the folder structure

```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ mv airflow/docker/prod docker/prod
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ rm -rf airflow
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ ls
airflow.sh          airflow_first_dags.pdf      dags                docker-compose.yaml
airflow_advanced_concepts.pdf  airflow_intro_and_install.pdf  data              logs
airflow_advanced_dags.pdf      airflow_subdags_sensors_xcom.pdf  docker            plugins

```

Rebuild production images

Transform image

```

ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ cd docker/prod/python_transform
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow/docker/prod/python_transform$ docker build -t python_transform .
[+] Building 0.9s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 216B
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load metadata for docker.io/library/python:3.8-slim
=> [1/6] FROM docker.io/library/python:3.8-slim@sha256:1d52838af602b4b5a831beb13a0e4d073280665ea7be7f69ce2382f29c5a61
=> [internal] load build context
=> => transferring context: 2.75kB
=> CACHED [2/6] WORKDIR /app
=> CACHED [3/6] COPY ./main.py ./main.py
=> CACHED [4/6] COPY ./requirements.txt ./requirements.txt
=> CACHED [5/6] RUN mkdir -p data/to_ingest
=> CACHED [6/6] RUN pip install -r requirements.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:4cf3a198f860a3558bc3818665a0344d4fc50ed1dda428c4f15de20cc4825d63
=> => naming to docker.io/library/python_transform:latest

```

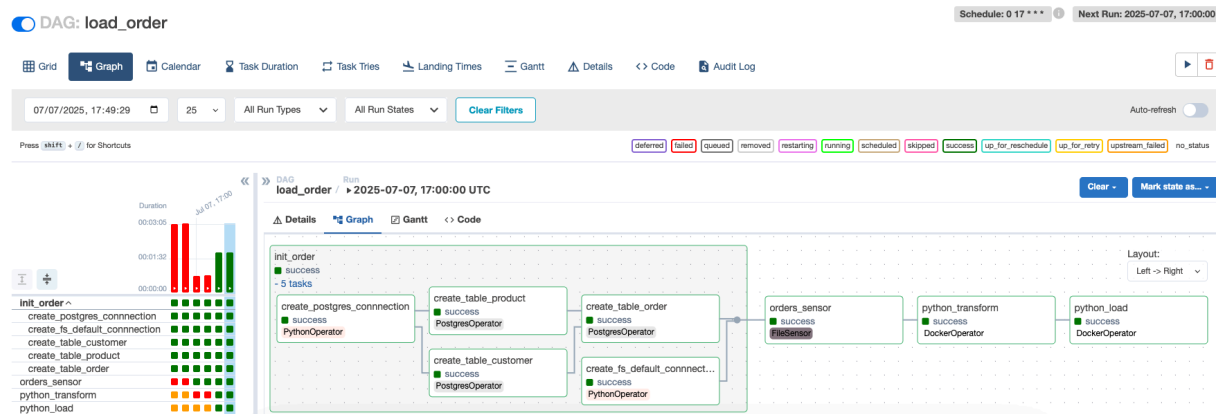
Add load_order.py file to airflow/dags. This DAG waits for orders.json to land, then runs two Docker containers in sequence. Transform JSON -> three files (orders.json -> orders.json, customers.csv, products.csv) and load those CSVs into Postgres.

Add plugins/build_init_order.py to airflow directory in order to publish init_order TaskGroup as a plugin

Reload new DAG

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker-compose restart airflow-scheduler airflow-worker
Restarting airflow_scheduler_1 ...
Restarting airflow_scheduler_1 ... done
Restarting airflow_worker_1 ... done
```

Open Airflow UI and trigger load_order DAG in Airflow UI



Verified product, order and customer appears in postgres

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker exec -it airflow_postgres_1 \
> psql -U airflow -d airflow -c "\dt"
```

List of relations			
Schema	Name	Type	Owner
public	ab_permission	table	airflow
public	ab_permission_view	table	airflow
public	ab_permission_view_role	table	airflow
public	ab_register_user	table	airflow
public	ab_role	table	airflow
public	ab_user	table	airflow
public	ab_user_role	table	airflow
public	ab_view_menu	table	airflow
public	alembic_version	table	airflow
public	callback_request	table	airflow
public	celery_taskmeta	table	airflow
public	celery_tasksetmeta	table	airflow
public	connection	table	airflow
public	customer	table	airflow
public	dag	table	airflow
public	dag_code	table	airflow
public	dag_owner_attributes	table	airflow
public	dag_pickle	table	airflow
public	dag_run	table	airflow
public	dag_run_note	table	airflow
public	dag_schedule_dataset_reference	table	airflow
public	dag_tag	table	airflow
public	dag_warning	table	airflow
public	dagrun_dataset_event	table	airflow
public	dataset	table	airflow
public	dataset_dag_run_queue	table	airflow
public	dataset_event	table	airflow
public	import_error	table	airflow
public	job	table	airflow
public	log	table	airflow
public	log_template	table	airflow
public	order	table	airflow
public	product	table	airflow
public	rendered_task_instance_fields	table	airflow
public	serialized_dag	table	airflow
public	session	table	airflow
public	sla_miss	table	airflow
public	slot_pool	table	airflow
public	task_fail	table	airflow

Verify that load_order has been successfully queued

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker exec -it airflow_scheduler_1 \
> airflow dags trigger load_order
/home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:812 DeprecationWarning: The sql_alchemy_conn
option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
/home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:724 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please update your config.
/home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:747 DeprecationWarning: The auth_backend option in [api] has been renamed to auth_backends - the old setting has been used, but please update your config.
/home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:761 FutureWarning: The auth_backends setting in [api] has had airflow.api.auth.backend.session added in the running config, which is needed by the UI. Please update your config before Apache Airflow 3.0.
/home/airflow/.local/lib/python3.8/site-packages/airflow/configuration.py:738 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
/home/airflow/.local/lib/python3.8/site-packages/airflow/settings.py:194 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
/home/airflow/.local/lib/python3.8/site-packages/airflow/models/base.py:71 DeprecationWarning: The sql_alchemy_conn option in [core] has been moved to the sql_alchemy_conn option in [database] - the old setting has been used, but please update your config.
[2025-07-07T17:53:36.557+0000] {_init_.py:42} INFO - Loaded API auth backend: airflow.api.auth.backend.basic_auth
[2025-07-07T17:53:36.560+0000] {_init_.py:42} INFO - Loaded API auth backend: airflow.api.auth.backend.session
```

conf	dag_id	dag_run_id	data_interval_start	data_interval_end	end_date	external_trigger	last_scheduling_decision	logical_date	run_type	start_date	state
{}	load_order	manual_2025-07-07T17:53:37+00:00	2025-07-06 17:00:00 +00:00	2025-07-07 17:00:00 +00:00	None	True	None	2025-07-07 17:53:37+00:00	manual	None	queued

Verify results on Postgres

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ for t in customer product order; do
> docker exec -it airflow_postgres_1 \
> psql -U airflow -d airflow \
> -c "SELECT count(*) FROM \"$t\";"
> done
count
-----
42
(1 row)

count
-----
135
(1 row)

count
-----
411
(1 row)
```

Spin up Adminer UI and connect to Postgres container

```
ubuntu@ip-172-31-46-112:~/datascientest-data-engineering/airflow$ docker run --network airflow_default -p 8081:8080 --
name adminer adminer
Unable to find image 'adminer:latest' locally
latest: Pulling from library/adminer
fe07684b16b8: Pull complete
0b94662290ef: Pull complete
498a67832465: Pull complete
e53785dc48fc: Pull complete
86c87a780257: Pull complete
f5aa21d09531: Pull complete
f3fccbf1255: Pull complete
481b52666b98: Pull complete
75a3724a01f7: Pull complete
75a5679ceb: Pull complete
a7970e51e658: Pull complete
04a922871959: Pull complete
4f4fb700ef54: Pull complete
c22d82ad8a15: Pull complete
e50c7ffe936a: Pull complete
049d5c30e22f: Pull complete
fe30f7e8173e: Pull complete
Digest: sha256:6bc678e77abcd8c7f34133f4903a4be485ec38b17df3d40a845ee56af0fcb32a
Status: Downloaded newer image for adminer:latest
[Mon Jul 7 17:59:45 2025] PHP 8.4.10 Development Server (http://[::]:8080) started
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22759 Accepted
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22700 Accepted
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22700 [200]: GET /
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22700 Closing
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22759 [200]: GET /?file=default.css&version=5.3.0
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22759 Closing
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22743 Accepted
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22743 [200]: GET /?file=dark.css&version=5.3.0
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22743 Closing
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22748 Accepted
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22748 [200]: GET /?file=functions.js&version=5.3.0
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22748 Closing
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22764 Accepted
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22764 [200]: GET /?file=logo.png&version=5.3.0
[Mon Jul 7 18:00:14 2025] [::ffff:95.90.232.82]:22764 Closing
[Mon Jul 7 18:00:15 2025] [::ffff:95.90.232.82]:22742 Accepted
[Mon Jul 7 18:00:15 2025] [::ffff:95.90.232.82]:22742 [200]: POST /?script=version
[Mon Jul 7 18:00:15 2025] [::ffff:95.90.232.82]:22742 Closing
```

Open URL <http://54.171.246.134:8081/> and fill in the details

Language: English

 **Adminer** 5.3.0

Login

SystemPostgreSQL

Serverpostgres

Usernameairflow

Password*****

Databaseairflow

Login

☐ Permanent login

Language: English PostgreSQL » postgres » airflow » Schema: public

 **Adminer** 5.3.0

db: airflow

Schemas: public

SQL command

Import

Export

Create table

select ab_permission

select ab_permission_view

select ab_permission_view_role

select ab_register_user

select ab_role

select ab_user

select ab_user_role

select ab_view_menu

select alembic_version

select callback_request

select celery_taskmeta

select connection

select customer

select dag

select dag_code

select dag_owner_attributes

select dag_pickle

select dag_run

select dag_run_note

select dag_schedule_dataset_refere

select dag_tag

select dag_warning

select dagrun_dataset_event

select dataset

select dataset_dag_run_queue

select dataset_event

select import_error

select job

select log

select log_template

select order

select product

select rendered_task_instance_field

Schema: public

Alter schema

Database schema

Tables and views

Search data in tables (47)

Table	Engine	Collation	Data Length ¹	Index Length ²	Data Free	Auto Increment	Rows ³	Comment ⁴
ab_permission	table		8,192	32,768	?	?	0	
ab_permission_view	table		40,960	32,768	?	?	~ 355	
ab_permission_view_role	table		40,960	32,768	?	?	~ 229	
ab_register_user	table		8,192	24,576	?	?	0	
ab_role	table		8,192	32,768	?	?	0	
ab_user	table		16,384	65,536	?	?	0	
ab_user_role	table		8,192	32,768	?	?	0	
ab_view_menu	table		8,192	32,768	?	?	~ 73	
alembic_version	table		8,192	16,384	?	?	0	
callback_request	table		8,192	8,192	?	?	0	
celery_taskmeta	table		303,104	229,376	?	?	~ 2,126	
celery_taskmeta	table		8,192	16,384	?	?	0	
connection	table		16,384	32,768	?	?	0	
customer	table		49,152	16,384	?	?	~ 42	
dag	table		96,304	49,152	?	?	~ 80	
dag_code	table		237,568	16,384	?	?	~ 65	
dag_owner_attributes	table		8,192	8,192	?	?	0	
dag_pickle	table		8,192	8,192	?	?	0	
dag_run	table		784,720	1,079,104	?	?	~ 800	

Selected (0)

Vacuum

Optimize

Truncate

Drop

Move to other database: public